

Unified Multimodal AI Platform for Automated E-Commerce Product Launch

1. Problem Statement

The rapid growth of e-commerce platforms has created significant opportunities for small businesses, local sellers, and home-based entrepreneurs to sell products online. However, launching a product on marketplaces such as Amazon, Flipkart, and Meesho requires multiple complex tasks including professional product photography, product demo videos, SEO-optimized product titles, attractive descriptions, packaging design, pricing analysis, category selection, and compliance verification. Most small sellers lack the technical expertise, design resources, and marketplace knowledge required to perform these tasks efficiently. Currently, sellers depend on multiple disconnected tools and manual workflows for image editing, video processing, content generation, pricing analysis, and packaging design. This process is time-consuming, costly, and inefficient, especially for businesses managing large inventories. Poor-quality product images, unoptimized videos, incomplete descriptions, incorrect categorization, and improper pricing often result in reduced product visibility, lower customer engagement, higher return rates, and decreased sales performance.

In real-world scenarios, marketplace sellers face several operational challenges such as:

- Creating professional product listings
- Generating engaging product videos and visuals
- Managing product categorization
- Estimating competitive pricing
- Understanding compliance requirements
- Scaling product onboarding efficiently
- Reducing dependency on external designers and content creators

Existing solutions generally focus on a single input modality such as text or images and fail to provide a unified platform capable of intelligently processing images, videos, speech, audio, and text together. Additionally, many systems lack scalable deployment, monitoring, automation, and production-ready MLOps infrastructure.

To address these challenges, there is a need for a Unified Multimodal AI Platform capable of accepting product images, videos, textual descriptions, and voice-based inputs to automatically generate marketplace-ready product assets. The system should integrate Computer Vision, Video Intelligence, Speech Recognition, OCR, Natural Language Processing, and Generative AI models while supporting scalable MLOps practices such as model deployment, monitoring, versioning, continuous integration, and retraining.

The proposed solution, "Unified Multimodal AI Platform for Automated E-Commerce Product Launch," aims to automate and simplify the entire product launch workflow by combining multimodal AI processing with scalable end-to-end MLOps infrastructure, thereby improving seller productivity, reducing manual effort, and accelerating product onboarding for e-commerce marketplaces.